



9512 - First Shift at an Elusive Dust Factory: Early dust and molecule formation in a giant star merger

Cycle: 4, Proposal Category: DD

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Viraj Karambelkar (PI)	Columbia University in the City of New York
Dr. Kishalay De (CoI)	Columbia University in the City of New York
Dr. Jacob Jencson (CoI)	California Institute of Technology
Dr. Ryan M Lau (CoI)	California Institute of Technology
Ms. Sam Rose (CoI)	California Institute of Technology
Dr. Nadejda Blagorodnova (CoI) (ESA Member)	Universitat de Barcelona
Dr. Marco Antonio Gomez-Munoz (CoI) (ESA Member)	Universitat de Barcelona

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	MIRI Imaging	MIRI Imaging	(1) WTP-M33-1
	2	LRS spectroscopy	MIRI Low Resolution Spectroscopy	(1) WTP-M33-1
	3	NIRSpec spectroscopy	NIRSpec Fixed Slit Spectroscopy	(1) WTP-M33-1

ABSTRACT

Eruptions from stellar mergers involving giant stars are predicted to be prolific dust factories contributing significantly to the cosmic dust budget. Yet these events have remained largely hidden for decades due to a lack of sensitive infrared searches. The only confirmed giant-star merger erupted over twenty years ago and was discovered only retrospectively in archival data. It was later found to have produced a dust yield rivaling core-collapse supernovae, but its early molecule and dust formation phases were entirely missed.

JWST Proposal 9512 (Created: Tuesday, November 18, 2025, 12:00:30PM Eastern Standard Time) - Overview

We have now discovered the second giant star merger in an ongoing search for mid-IR explosions using the NEOWISE survey. Unlike its predecessor, this source has been caught at an early phase, with recent mid-IR photometry indicating that dust formation has just begun. This presents an unprecedented opportunity to study early molecular and dust formation in a highly dust-productive environment.

Here, we propose a non-disruptive JWST DD observation of this source, combining NIRSpec to characterize molecules such as CO, SiO, water-vapor, and potentially water-ice, MIRI LRS spectroscopy to detect the seeds of oxygen-rich dust, and long-wavelength MIRI imaging to trace the coldest dust components. These observations will provide the earliest snapshot of silicate dust formation in a giant star merger, and test models of oxygen-rich dust formation and common-envelope evolution. With dust formation already underway, deferring to Cycle 5 risks the source becoming too faint for spectroscopy and entirely missing out on the early dust-forming phase of this likely once-in-a-JWST-lifetime event.

OBSERVING DESCRIPTION

In this proposal, we aim to observe a newly discovered giant star merger WTP-M33-1 - the first in the last two decades and only the second one overall - with the James Webb Space Telescope to characterize its early molecular and dust content. Our observations constitute a total time of 4.9 hours including overheads (1.9 hours science), and comprise -

- 1) NIRSpec spectroscopy : With the high-resolution G235H/F170LP (2828 seconds) and G395H/F290LP (1514 seconds) grating-filter combinations covering 1.6 to 5um
- 2) MIRI LRS spectroscopy covering 5-12um (1676 seconds)
- 3) MIRI imaging : In F1280W, F1500W, F1800W, F2100W (122 seconds each), and F2550W (255 seconds) filters.

Proposal 9512 - Targets - First Shift at an Elusive Dust Factory: Early dust and molecule formation in a giant star merger

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	WTP-M33-1	RA: 01 33 18.9729 (23.3290537d) Dec: +30 52 13.23 (30.87034d) Equinox: J2000	Epoch of Position: 2000.0	
Comments: Category=Star Description=[M stars, Nova-like variable stars, Novae] Extended=NO					

Proposal 9512 - Observation 1 - First Shift at an Elusive Dust Factory: Early dust and molecule formation in a giant star merger

Observation	Proposal 9512, Observation 1: MIRI Imaging										Tue Nov 18 17:00:30 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	WTP-M33-1	RA: 01 33 18.9729 (23.3290537d)			Epoch of Position: 2000.0					
			Dec: +30 52 13.23 (30.87034d)								
			Equinox: J2000								
Template	Comments: Category=Star Description=[M stars, Nova-like variable stars, Novae] Extended=NO										
	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		1	1			DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1280W	FASTR1	5	2	1	Dither 1	4	8	122.102	
	2	F1500W	FASTR1	5	2	1	Dither 1	4	8	122.102	
	3	F1800W	FASTR1	5	2	1	Dither 1	4	8	122.102	
	4	F2100W	FASTR1	5	2	1	Dither 1	4	8	122.102	
	5	F2550W	FASTR1	5	4	1	Dither 1	4	16	255.304	
Special Requirements	Between Dates 08-DEC-2025:00:00:00 and 29-JAN-2026:00:00:00										
	Group Observations 1, 2, 3, Non-interruptible										

Proposal 9512 - Observation 2 - First Shift at an Elusive Dust Factory: Early dust and molecule formation in a giant star merger

Observation	Proposal 9512, Observation 2: LRS spectroscopy									Tue Nov 18 17:00:30 GMT 2025
	Diagnostic Status: Warning									
Diagnostics	Observing Template: MIRI Low Resolution Spectroscopy									
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(1)	WTP-M33-1	RA: 01 33 18.9729 (23.3290537d)			Epoch of Position: 2000.0				
			Dec: +30 52 13.23 (30.87034d)							
			Equinox: J2000							
	Comments: Category=Star Description=[M stars, Nova-like variable stars, Novae] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	F560W	FAST	4	1	1	11.1	267806	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	5	1	1	1	1	13.875		F560W

Proposal 9512 - Observation 2 - First Shift at an Elusive Dust Factory: Early dust and molecule formation in a giant star merger

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	100	3	6	1	2	1676.124	
Special Requirements	Between Dates 08-DEC-2025:00:00:00 and 29-JAN-2026:00:00:00								
	Group Observations 1, 2, 3, Non-interruptible								

Proposal 9512 - Observation 3 - First Shift at an Elusive Dust Factory: Early dust and molecule formation in a giant star merger

Observation	Proposal 9512, Observation 3: NIRSpec spectroscopy										Tue Nov 18 17:00:30 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	WTP-M33-1	RA: 01 33 18.9729 (23.3290537d)			Epoch of Position: 2000.0						
			Dec: +30 52 13.23 (30.87034d)									
			Equinox: J2000									
		Comments: Category=Star Description=[M stars, Nova-like variable stars, Novae] Extended=NO										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	1 WTP-M33-1	WATA	FULL	F110W	NRSRAPID	3	1	1	42.947	267806	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	S200A1	NRSIRS2RAPID	35	2	1	NONE	3	6	3151.2	
	2	G395H/F290LP	S200A1	NRSIRS2RAPID	40	1	2	NONE	3	3	1794.433	

Proposal 9512 - Observation 3 - First Shift at an Elusive Dust Factory: Early dust and molecule formation in a giant star merger

Special Requirements	Between Dates 08-DEC-2025:00:00:00 and 29-JAN-2026:00:00:00 Group Observations 1, 2, 3, Non-interruptible
----------------------	---